

Programme

Diploma in Software Development – 240 Credits
(DSD)
and
Bachelor of Software Engineering – 360 credits
(BSE)

Year Two

Course

202 Cross-Platform Development
(Level 6, 15 Credits, Version 2)

Assessment 2

Research Proposal

Weighting within course:

75%

Objective

This course aims to students with further understanding of academic and professional investigation and presentation approaches, in order to creatively ideate and apply critical thinking to the production and presentation of well-researched and persuasive proposals for software products.

The objective of this assessment is to help you practice proposing an innovative Research & Development (R&D) project focused on creating high-performing, cross-platform web apps. By considering project goals, target audiences, and available technologies for different operating systems, you will develop vital skills in collaborative teamwork, thorough research, and effective project proposal writing. This assessment aims to build your confidence in evaluating and selecting technical options through a combination of research, teamwork, and potential field reviews, while also enhancing communication skills and the ability to make informed decisions in dynamic professional environments.

Learning Outcomes (LOs) covered

- LO 1** Integrate an understanding of academic practice and research methods to identify and investigate subjects of interest in relation to professional projects and directions.
- LO 2** Generate effective communication to share ideas and concepts and enhance professional practice.
- LO 3** Demonstrate understanding of how development in an area of personal expertise can be used to contribute to the execution of a team-based software project.
- LO 4** Integrate investigative findings to develop a coherent argument for a project proposal.

DSD Qualification Graduate Profile Outcomes (GPOs) covered

- GPO 1** Analyse requirements and apply appropriate modelling tools, SDLC and HCI principles, to design and document software solutions for a range of problems in an organisational context, including creating accurate and clear technical and user documentation/resources
- GPO 2** Write and maintain complex programs using design patterns, data structures and algorithms to meet specifications and software development standards.
- GPO 3** Apply a range of software quality assurance techniques, including designing and executing a variety of tests, to verify correctness of systems on a range of platforms to meet requirements, and document results.
- GPO 5** Integrate security and privacy principles throughout software development to ensure application security and system integrity using current best practices.
- GPO 8** Apply professional and ethical practices with integrity to meet the industry wide expectations of a responsible IT professional, in accordance with legal, regulatory and organisational requirements.
- GPO 9** Apply communication, information design, teamwork, personal, and interpersonal skills, to enhance working effectiveness, efficiency, and quality outcomes in a variety of situations in an organisational environment.
- GPO 10** Apply project management tools and techniques, using critical thinking, enterprise skills and knowledge of project planning, management and control, to an IT related project, to analyse and solve problems.

BSE Qualification Graduate Profile Outcomes (GPOs) covered

- GPO 1** Employ critical understanding of business and management practices for professional software projects and environments.
- GPO 2** Apply a range of interpersonal, professional and academic communication approaches to optimise opportunities, relationships and outcomes with stakeholders.
- GPO 4** Identify and fulfill self-learning needs through proactive engagement in a range of approaches to personal and professional development, including disciplined self-reflection and soliciting the views of others.
- GPO 5** Exercise imaginative entrepreneurial skills to devise creative and viable software concepts, through independent and collective approaches to idea generation.
- GPO 6** Conduct transdisciplinary investigation and analysis to enhance the production of user-centric, viable, ethical and socially responsive creative software solutions.
- GPO 7** Employ solution-focused design-thinking approaches to contribute to the ideation, development and production of software.
- GPO 8** Analyse, articulate, and apply theories informing personal practice within the software industry.
- GPO 9** Integrate an adaptive set of specialist skills, approaches, and tools with a holistic understanding of software development practices, to produce software within transdisciplinary teams.

Assessment Matrix

Learning Outcome	GPO (DSD)	GPO (BSE)	Assessment Task	Weightage (%)
LO 1: Integrate an understanding of academic practice and research methods to identify and investigate subjects of interest in relation to professional projects and directions.	GPO 1	GPO 1	Product concept (including rationale)	10
	GPO 2	GPO 2	(LO4)	
	GPO 3	GPO 4	Field review	15
	GPO 5	GPO 5	(LO2)	
LO 2: Generate effective communication to share ideas and concepts and enhance professional practice.	GPO 8	GPO 6		
	GPO 9	GPO 7	Production plan (including proposed roles and technologies to be used)	25
LO 3: Demonstrate understanding of how development in an area of personal expertise can be used to contribute to the execution of a team-based software project.	GPO 10	GPO 8 GPO 9	(LO2, LO3)	
			References (LO1)	5
			Research output (LO4)	25
			Presentation pitch (LO2)	10
LO 4: Integrate investigative findings to develop a coherent argument for a project proposal.			Individual reflection (LO3)	10
Total				100

Grading

The final grade will be determined by the score achieved in this assessment based on the following table. Should a second or third attempt be required the maximum contribution toward the overall mark for the tasks that required a second or third assessment attempt is 50%. A late submission is considered a second attempt, so the contribution will be capped at 50%.

Grade	Range
A +	Meet all course requirements, range (90—100%)
A	Meet all course requirements, range (85—89%)
A -	Meet all course requirements, range (80—84%)
B +	Meet all course requirements, range (75—79%)
B	Meet all course requirements, range (70—74%)
B -	Meet all course requirements, range (65—69%)
C +	Meet all course requirements, range (60—64%)
C	Meet all course requirements, range (55—59%)
C -	Meet all course requirements, range (50—54%)
D	Did not meet all course requirements, range (40—49%)
E	Did not meet all course requirements, mark range (0—39%)

Candidate's Assessment Instructions

- This assessment is an **open-book activity**, you can use your own course and review notes as well as offline or online resources, such as textbooks or online journals.
- Teams with high degrees of social sensitivity are known to perform well when executing several specialised collaborative activities. In relation to this, you need to collaborate with your peers to form a team of 2-3 people and your team needs to conduct research for a hypothetical client.
- You will each submit an identical piece of work separately for the work that requires collaboration.
- You can always ask your online tutor if you need further explanation about forming a group or if the instructions are unclear.
- Your work should not be plagiarised. Plagiarism includes copying material without acknowledging it, copying from another student, getting another person to help you with your assessment, using material from commercial essays or assignment services, or using AI to create the answers.
- The purpose of this assessment is to assess your knowledge. In the event Yoobee suspects collusion, this will be addressed. For more information on plagiarism, please refer to the Student Handbook.
- Submit your completed assessment online in the correct space provided.
- Marks and feedback will be returned within 15 days of the submission date.
- By completing and submitting an assessment you are authenticating that the work is original and does not violate plagiarism or copyright law. Authenticity is checked where any breaches of academic integrity are suspected. Please refer to the Student Handbook for further information.

Submission Instructions

Please submit the following to your LMS:

- Your team research project proposal, including all team member names, student IDs, and course codes.
- Presentation file
- Individual reflection report from every team member

Assessment Tasks Completion Timeline

Week	What needs to be done	What needs to be submitted
Week1	Assessments 1 and 2 to be released. Student starts working on Assessment 1	
Week 2	Student keeps working on Assessment 1	
Week 3	Recommend completing Assessment 1 by the end of the week	
Week 4	Students for a group. The group starts working on Assessment 2	Submit via LMS: Assessment 1 document
Week 5	The group continues working on Assessment 2	
Week 6	The group continues working on Assessment 2	
Week 7	The group continues working on Assessment 2	
Week 8	Student to complete Assessment 2 with individual reflection	Submit via LMS: - Your team project proposal, including all team member names, student IDs, and course codes. - Presentation file - Individual reflection report from every team member

ASSESSMENT

LO 1: Integrate an understanding of academic practice and research methods to identify and investigate subjects of interest in relation to professional projects and directions.

LO 2: Generate effective communication to share ideas and concepts and enhance professional practice.

LO 3: Demonstrate understanding of how development in an area of personal expertise can be used to contribute to the execution of a team-based software project.

LO 4: Integrate investigative findings to develop a coherent argument for a project proposal.

PART A

For this part of the assessment, you would need to form a team of two to three students (preferably two).

You need to work together to write a proposal and present it.

Assessment Brief

As the head of the R&D department, you are responsible for generating and evaluating concepts for your company's upcoming project. The project involves developing a globally accessible web application through your online platform, offering real-time functionality. This application must deliver consistent performance across different browsers on both desktop computers and mobile devices, ensuring a uniform and exceptional user experience.

Assessment Tasks

Tasks

Write a detailed project proposal containing a production plan, your research output, and design prototypes.

Include sections in your proposal according to the headings and details below:

Title Page

- Course and assessment details: 202 Cross-Platform Development I Assessment 2- Research Proposal
- Your name
- Your student ID
- Class code

Product concept (including rationale)

This is a simple summary (250 to 500 words) of the product concept and a brief overview of the features provided by the product you are proposing. Answer questions about the product like:

- Product rationale
- Proposed features
- Programming framework/technology selected.

Field review

If field review is being used, it will include the following:

- What is the existing system (if any) in place to meet the business objectives?
- What are the problems being faced by the users of the existing system (for example, in terms of UX/UI)?
- What are the features desirable by the existing users in an improved or new system?
- What are the proposed features of the new system (for example, improved UX/UI, online booking, payment system, voice user interface, accessibility options etc.)

In the context of creating a novel system that doesn't involve enhancing an existing one, you have the opportunity to develop an innovative concept. In this scenario, you can delve into case studies of analogous systems or draw inspiration from existing literature. Employing a comparison and contrast methodology, you can assess the merits of your proposed system against these comparable solutions within the relevant domain.

Production plan

This can include the following:

- Project duration and chosen project methodology (Agile, Waterfall, etc.)
- Project phases and work breakdown structure (WBS)
- Gantt chart (including milestones/project deliverables, dependencies etc)
- Communication plan
- Proposed roles (for example, if Scrum has been adopted, then you can talk about the roles of product owner, Scrum master, and developers).
- Project management tools (scheduling, version control, etc.)

References

You are encouraged to survey at least five relevant conference/journal articles in the proposed area of research/investigation and should summarize them in the research output section of the report. All the literature review performed should be properly referenced in APA Version 7 format.

Include a complete list of references used in the report at the end of the report under REFERENCES section.

Note: Please use EBSCO database provided by college. You can also use Google Scholar to find articles. Reference management systems like ReadCube, Zotero, EndNote, Mendeley can be used to organize all your references in one place. Additionally, MS Word's referencing features can be used to edit your references.

Research Output

This proposed research output will include (but is not limited to) the following:

- At least 5 relevant conference papers or journal articles should be summarized in this section and proper citations in APA Version 7 to the published work should be correctly provided.
- Three user personas for the proposed system.
- Use case diagrams and UML sequence diagrams.
- Mock-ups and digital Lo-Fi/Mid-Fi wireframes of the proposed UI (responsive webpages).
- Usability Testing of the Lo-Fi wireframes with at least five users.

Presentation

Present your team project proposal to your class. Cover each of the above headings. Presentations should be no longer than 15 minutes.

PART B

This part of the assessment is individual work, and you are required to complete it on your own.

Individual Reflection

The individual reflection report should be submitted as a written document or a link to the personal blog and will include your thoughts about your contribution to the following:

- Investigation/research.
- UI prototyping.
- Team formation.
- Personal learning outcomes.

Marking Rubrics

To pass this assessment, you must meet all the requirements of each of the learning outcomes (irrespective of the numerical grade awarded) and achieve PASS results for each learning outcome.

Criterion and Weightage		Evidence				
Task		A+, A, A-	B+, B, B-	C+, C, C-	D	E
LO 1, 2, 3 and 4 (DSD: GPO 1, GPO 2, GPO 3, GPO 5, GPO 8, GPO 9, GPO 10 – BSE: GPO 1, GPO 2, GPO 4, GPO 5, GPO 6, GPO 7, GPO 8, GPO 9)						
Product concept (including rationale) (LO4)	10%	All the required elements in the initial product concept have been described correctly and with full clarity.	All the required elements in the initial product concept have been described with less clarity.	The required elements in the initial product concept have been described but some details are missing.	The required elements in the initial product concept have been described with ambiguity or NOT described at all.	Mostly irrelevant evidence. OR Task not attempted.
Field review (LO2)	15%	All the required elements in the field review have been fully described and with full clarity or two relevant case studies have been described.	All the required elements in the field review have been described but with less clarity or one relevant case study has been described.	All the required elements in the areas of investigation have been described but some details are missing, or one case study has been discussed but lacks details.	All the required elements in the field review have been described with ambiguity or NOT described at all or no case study has been discussed	Mostly irrelevant evidence. OR Task not attempted.
Production plan (including proposed roles and technologies to be used) (LO2, LO3)	25%	All the required elements in the production plan have been described correctly and with full clarity.	All the required elements in the production plan have been described with less clarity.	The required elements in the production plan have been described but some details are missing.	The required elements in the production plan have been described with ambiguity or NOT described at all.	Mostly irrelevant evidence. OR Task not attempted.
References (LO1)	5%	All the references have been edited in the APA Version 7 format correctly.	Five references have been edited in the APA Version 7 format correctly.	Three references have been edited in the APA Version 7 format correctly.	None of the references have been edited in the APA Version 7 format correctly.	Mostly irrelevant evidence. OR Task not attempted.
Research output (LO4)	25%	All the required elements in the research output have been included, described correctly and with full clarity.	All the required elements in the research output have been included, described but with less clarity.	The required elements in the research output have been included, described but some details are missing.	The required elements in the research output have been included, described with ambiguity or NOT described at all.	Mostly irrelevant evidence. OR Task not attempted.
Presentation pitch (LO2)	10%	All the required elements in the PowerPoint presentation have been included/described with clarity and the presentation is completed in the allotted time.	All the required elements in the PowerPoint presentation have been included/described with less clarity and the presentation is completed in the allotted time.	All the required elements in the PowerPoint presentation have been included/described with some missing details and the presentation is completed in the allotted time.	Mostly irrelevant evidence.	Task not attempted.
Individual reflection (LO3)	10%	All the required elements in the individual reflection have been described correctly and with full clarity.	All the required elements in the individual reflection have been described with less clarity.	The required elements in the individual reflection have been described but some details are missing.	All the required elements in the PowerPoint presentation have been included/described with ambiguity or are not included/described at all and the presentation is not completed in the allotted time.	Mostly irrelevant evidence. OR Task not attempted.
Sum	100%					

